DATA-DRIVEN OPTIMIZATION OF IT SUPPORT TEAM PERFORMANCE USING ANALYTICS

Project Report

Name: Shaik Farhana

Internship: Infosys Internship

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1. Introduction

This project focuses on analyzing IT support ticket data to improve the overall performance and efficiency of the support team. In today's organizations, customer support plays a crucial role in maintaining user satisfaction and resolving issues quickly.

Using Power BI, various data visualization techniques were applied to understand ticket trends, resolution times, and customer satisfaction levels. The analysis helps in identifying performance gaps, high-priority issues, and areas that require improvement.

The main objective of this project is to use data-driven insights to support better decision-making and enhance the effectiveness of IT support services.

2.Dataset Description

The dataset used in this project contains information related to IT support tickets. It includes multiple attributes that help in analyzing different aspects of ticket handling and performance.

The key columns in the dataset are:

- Ticket ID: Unique identifier for each support ticket

- Ticket Type: Category of the issue (e.g., technical, billing, cancellation)

- Ticket Priority: Indicates urgency level (Low, Medium, High, Critical)

- Ticket Channel: Source of the ticket (Phone, Email, Chat, Social Media)

- Resolution Duration: Time taken to resolve the ticket

- Customer Satisfaction Rating: Feedback score provided by users

This dataset enables detailed analysis of ticket distribution, resolution efficiency, and customer experience. It serves as the foundation for creating meaningful insights and visualizations in the project.

3.Modules Implemented

- **Module 1: Project Initialization and Dataset Setup**
- **Module 2: Data Cleaning and Feature Engineering**
- **Module 3: Exploratory Visualization**
- **Module 4: Similarity and Cluster Insights**
- **Module 5: Performance Analysis**
- **Module 6: Geographic Insights**
- **Module 7: Dashboard Preparation**

4.Visualizations and Analysis

Figure 1: Ticket Type Distribution

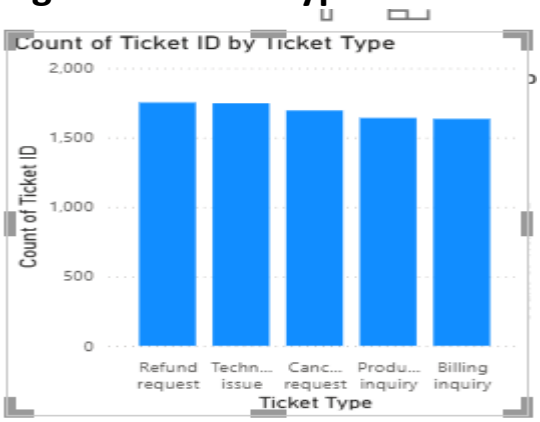


Figure 2: Average Resolution Time by Priority

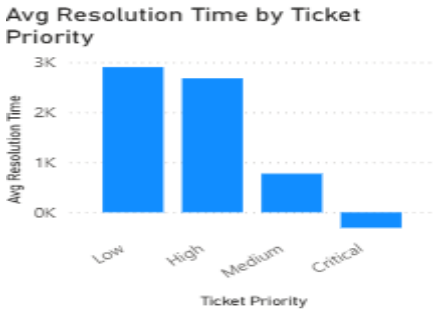


Figure 3: Tickets by Channel

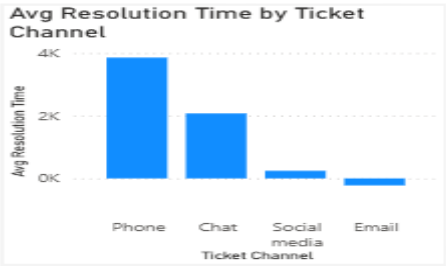


Figure 4: Cluster Analysis

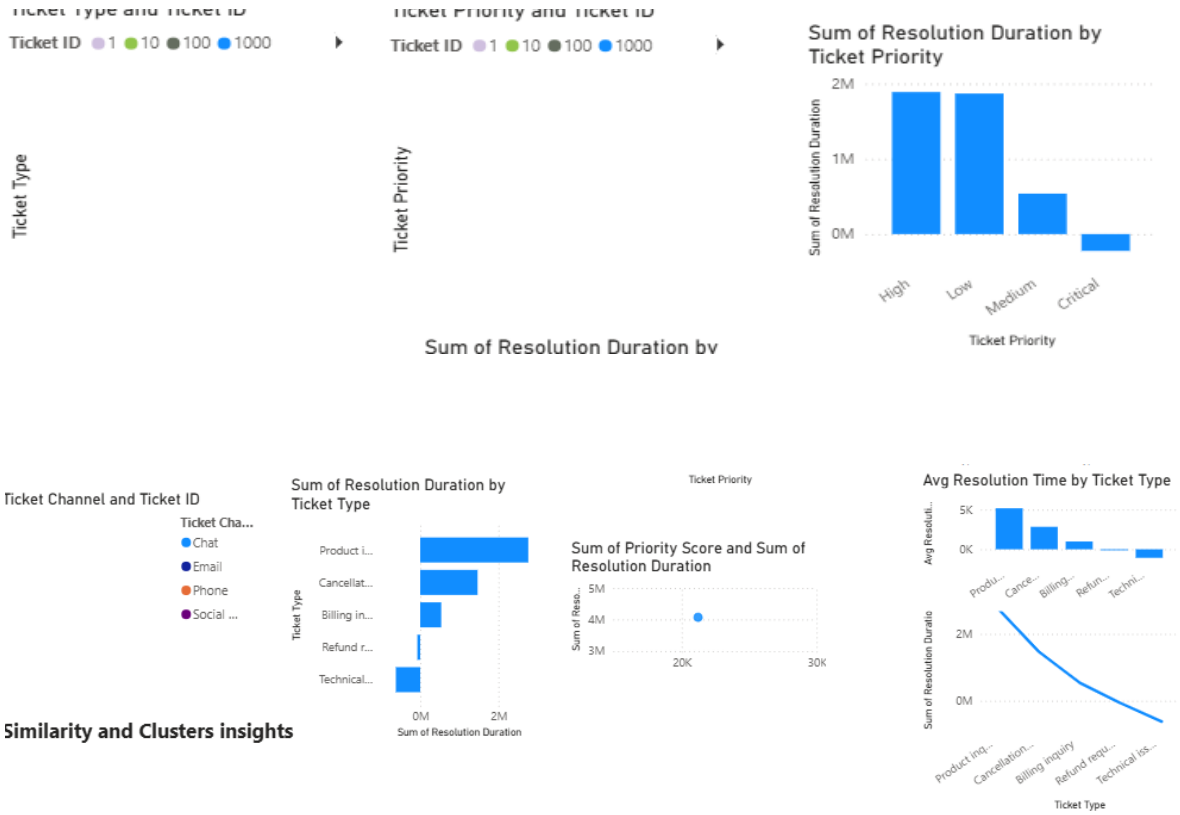


Figure 5: Resolution Duration by Ticket Type

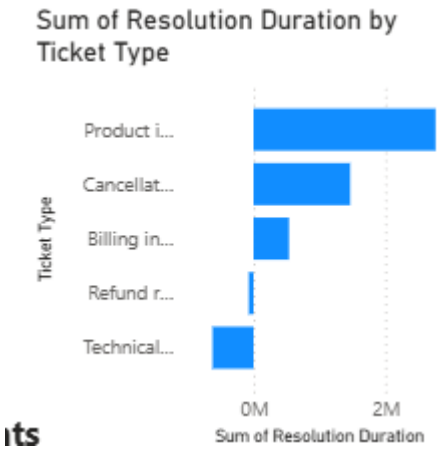


Figure 6: Total Tickets Overview

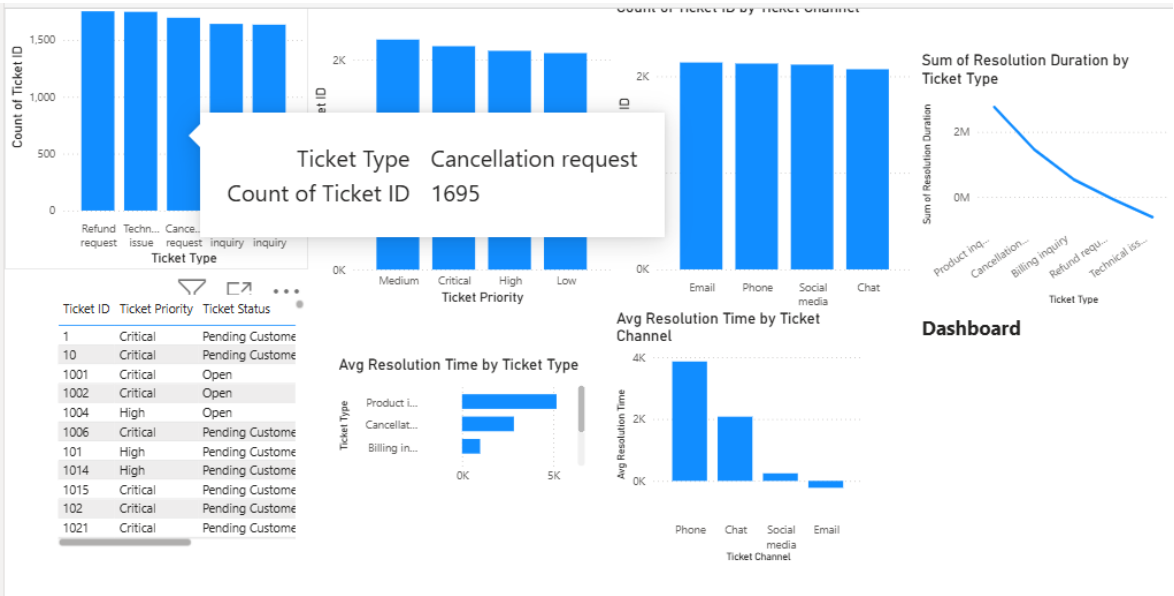
This KPI shows the total number of tickets handled in the dataset.

8469

Total Tickets

Figure :7 — Final Dashboard

This dashboard provides a complete overview of ticket performance, resolution time, and customer insights.



5.Key Insights

- **High priority tickets take more time**
- **Phone and chat channels handle more tickets**
- **Some ticket types are more frequent**
- **Customer satisfaction varies with priority**

The insights gained from the dashboard can help organizations identify problem areas, optimize resource allocation, and enhance overall customer satisfaction.

6.Conclusion

This project demonstrates how data analysis can significantly improve the performance of an IT support team. By analyzing ticket data using Power BI, important insights were identified regarding resolution time, ticket priority, and customer satisfaction.

The dashboard helps in understanding which types of issues occur frequently, how efficiently they are resolved, and which channels handle more tickets. It also highlights high-priority tickets that require immediate attention.

Overall, this project shows that data-driven decision-making can help organizations optimize their support processes, reduce resolution time, and improve customer satisfaction. With further enhancements, such as real-time data integration and advanced analytics, the system can become even more effective in supporting business operations.